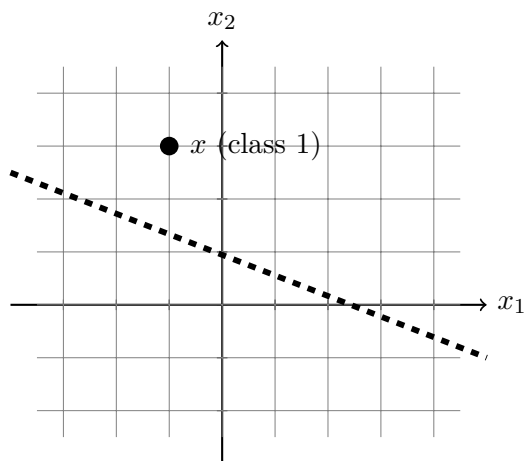


Name: _____

Student ID: _____

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.